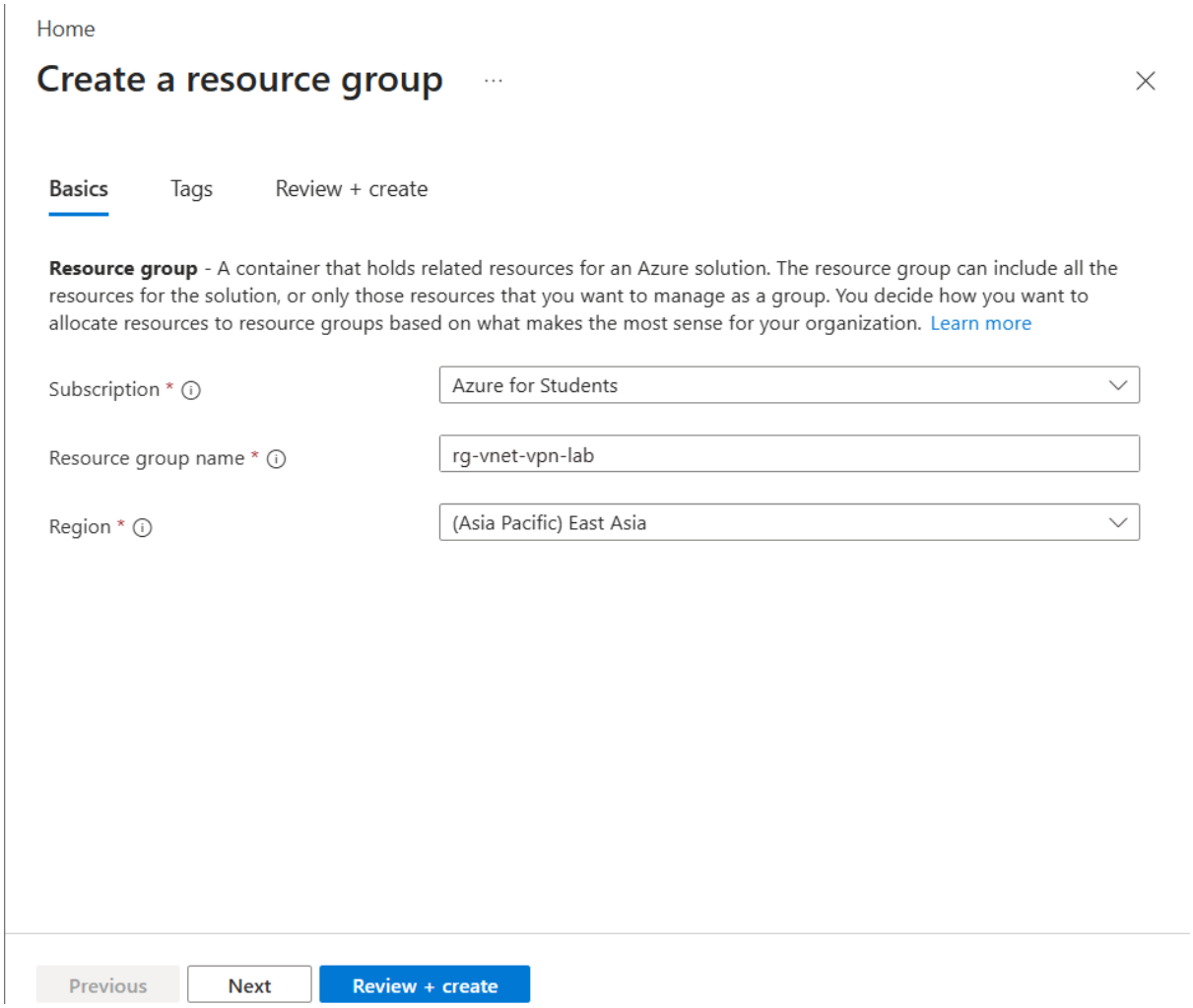
# ASSESSMENT-14

**Step 1: Create Resource Group and always use the same region as the resorceg**




**Step 2: Create Virtual Network and their respective Subnet and Gateway Subnet**

**a.VNET-1**

Home > Network foundation | Virtual networks

# Create virtual network

Basics | Security | Address space | Tags | Review + create

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * : Azure for Students

Resource group * : rg-vnet-vpn-lab

Create new

## Instance details

Virtual network name * : vnet1

Region * : (Asia Pacific) East Asia

Deploy to an Azure Extended Zone

Validation passed

Basics | Security | Address space | Tags | Review + create

| | |
|---|---|
| Name | vnet1 |
| Region | East Asia |

**Security**

| | |
|---|---|
| Azure Bastion | Disabled |
| Azure Firewall | Disabled |
| Azure DDoS Network Protection | Disabled |

**Address space**

| | |
|---|---|
| Address space | 10.0.0.0/16 (65,536 addresses) |
| Subnet | default (10.0.0.0/24) (256 addresses) |
| - Private subnet | Enabled |

**Tags**

# Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. Learn more

Subnet purpose: Default

Name *: subnet1

**IPv4**

Include an IPv4 address space: ☑

IPv4 address range: 10.0.0.0/16

10.0.0.0 - 10.0.255.255

Starting address *: 10.0.1.0

Size: /24 (256 addresses)

Subnet address range: 10.0.1.0 - 10.0.1.255

**IPv6**

Include an IPv6 address space: ☐ This virtual network has no IPv6 address ranges.

**Private subnet**

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual

Add | Cancel | Give feedback

# Add a subnet

**Adding subnet**

Adding subnet 'GatewaySubnet' to virtual network 'vnet1'.

Select an address space and configure your subnet. You can customiz
you plan to add select services later. Learn more

Subnet purpose: Virtual Network Gateway

Name *: GatewaySubnet

**IPv4**

Include an IPv4 address space: ☑

IPv4 address range: 10.0.0.0/16

10.0.0.0 - 10.0.255.255

Starting address *: 10.0.255.0

Size: /27 (32 addresses)

Subnet address range: 10.0.255.0 - 10.0.255.31

**IPv6**

Include an IPv6 address space: ☐ This virtual network has no IPv6 address ranges.

**Private subnet**

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual

Add | Cancel | Give feedback

Microsoft Azu | Search resources, services, and docs (G+/) | Copilot

Home > vnet1-1775882923835 | Overview > vnet1

# vnet1 | Subnets

Virtual network

**Successfully added subnet**

Successfully added subnet 'GatewaySubnet' to virtual network 'vnet1'.

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
  - Address space
  - Connected devices
  - Subnets
  - Bastion
  - DDoS protection
  - Firewall
  - Microsoft Defender for Cloud
  - Network manager

+ Subnet | Refresh | Manage users | Delete

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

| Name ↑ | IPv4 | IPv6 | Available IP |
|---|---|---|---|
| default | 10.0.0.0/24 | - | 251 |
| subnet1 | 10.0.1.0/24 | - | 251 |
| GatewaySu... | 10.0.255.0/27 | - | availability |

Showing 3 subnets

Give feedback

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * : Azure for Students

Resource group * : rg-vnet-vpn-lab

Create new

## Instance details

Virtual network name * : vnet2

Region * : (Asia Pacific) East Asia

Deploy to an Azure Extended Zone

**b.VNET-2**

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students

Resource group * rg-vnet-vpn-lab

Create new

**Instance details**

Virtual network name * vnet2

Region * (Asia Pacific) East Asia

Deploy to an Azure Extended Zone

Home > Network foundation | Virtual networks

# Create virtual network

Basics Security Address space Tags Review + create

Allocate using IP address pools. Learn more

Add a subnet

10.1.0.0/16 Delete address space

10.1.0.0 /16

10.1.0.0 - 10.1.255.255 65,536 addresses

| Subnets | IP address range | Size | NAT gateway |
|---|---|---|---|
| default | 10.1.0.0 - 10.1.0.255 | /24 (256 addresses) | - |

Add IPv4 address space

Previous Next Review + create Give feedback

## Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. Learn more

Subnet purpose: Virtual Network Gateway

Name *: GatewaySubnet

**IPv4**

Include an IPv4 address space: ☑

IPv4 address range: 10.1.0.0/16

10.1.0.0 - 10.1.255.255

Starting address *: 10.1.255.0

Size: /27 (32 addresses)

Subnet address range: 10.1.255.0 - 10.1.255.31

**IPv6**

Include an IPv6 address space: ☐ This virtual network has no IPv6 address ranges.

**Private subnet**

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual

Add | Cancel | Give feedback

Microsoft Azu | Search resources, services, and docs (G+/) | Copilot

## Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. Learn more

Subnet purpose: Default

Name *: subnet1

**IPv4**

Include an IPv4 address space: ☑

IPv4 address range: 10.1.0.0/16

10.1.0.0 - 10.1.255.255

Starting address *: 10.1.1.0

Size: /24 (256 addresses)

Subnet address range: 10.1.1.0 - 10.1.1.255

**IPv6**

Include an IPv6 address space: ☐ This virtual network has no IPv6 address ranges.

**Private subnet**

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual

Add | Cancel | Give feedback

## Step 3: Create VPN Gateway for VNet-1 and VNET-2

Home > Create a resource > Marketplace

# Create virtual network gateway ...

Validation passed

Basics Tags **Review + create**

**Basics**

| | |
|---|---|
| Subscription | Azure for Students |
| Resource group | rg-vnet-vpn-lab |
| Name | vnet1-gateway |
| Region | East Asia |
| SKU | VpnGw1AZ |
| Generation | Generation1 |
| Virtual network | vnet1 |
| Subnet | GatewaySubnet (10.0.255.0/27) |
| Gateway type | Vpn |
| VPN type | RouteBased |
| Enable active-active mode | Disabled |
| Enable Advanced Connectivity | Disabled |
| Configure BGP | Enabled |
| Autonomous system number (ASN) | 65515 |

Create Previous Next Download a template for automation

Home > Create a resource > Marketplace

# Create virtual network gateway ...

Subscription * Azure for Students

Resource group rg-vnet-vpn-lab (derived from virtual network's resource group)

**Instance details**

Name * vnet2-gateway

Region * East Asia

Deploy to an Azure Extended Zone

**Gateway Configuration**

Gateway type * VPN ExpressRoute

SKU * VpnGw1AZ

Generation Generation1

Enable Advanced Connectivity Enabled Disabled

Virtual network * vnet2

Create virtual network

**Step 4: Create Connection (VNet1 → VNet2)**

Home › vnet1-gateway | Connections

# Create connection ...

Validation passed

Basics Settings Tags **Review + create**

## Basics

| | |
|---|---|
| Subscription | Azure for Students |
| Resource Group | rg-vnet-vpn-lab |
| Region | East Asia |
| Connection type | VNet-to-VNet |
| Connection name | vnet1-to-vnet2 |

## Settings

| | |
|---|---|
| First virtual network gateway | vnet1-gateway |
| Second virtual network gateway | vnet2-gateway |
| IKE Protocol | IKEv2 |
| IPsec / IKE policy | Default |
| Use policy based traffic selector | Disable |
| DPD timeout in seconds | 45 |
| Connection Mode | Default |

Create Previous Next Download a template for automation

**Step 5: Create Reverse Connection (VNet2 → VNet1)**

Home › vnet2-gateway | Connections

# Create connection

Validation passed

Basics Settings Tags Review + create

## Basics

| | |
|---|---|
| Subscription | Azure for Students |
| Resource Group | rg-vnet-vpn-lab |
| Region | East Asia |
| Connection type | VNet-to-VNet |
| Connection name | vnet2-to-vnet1 |

## Settings

| | |
|---|---|
| First virtual network gateway | vnet2-gateway |
| Second virtual network gateway | vnet1-gateway |
| IKE Protocol | IKEv2 |
| IPsec / IKE policy | Default |
| Use policy based traffic selector | Disable |
| DPD timeout in seconds | 45 |
| Connection Mode | Default |

Create Previous Next Download a template for automation

Microsoft Azu | Search resources, services, and docs (G+/) | Copilot

Home

# CreateConnection-20260411112636 | Overview

Deployment

Search

Overview
Inputs
Outputs
Template

Delete | Cancel | Redeploy | Download | Refresh

## Your deployment is complete

Deployment name : CreateConnection-20260411112636
Subscription : Azure for Students
Resource group : rg-vnet-vpn-lab
Start time : 11/04/2026, 11:27:34
Correlation ID : 73267381-7b74-48c1-a26e-0b5fa9d9e627

Deployment details

Next steps

Go to resource

**Cost management**
Get notified to stay within your budget and prevent unexpected charges on your bill.

*Add or remove favorites by pressing Ctrl+Shift+F*

Home

# CreateVm-canonical.ubuntu-24_04-lts-server-2026041...

Deployment

Search

Overview
Inputs
Outputs
Template

Delete | Cancel | Redeploy | Download | Refresh

## Your deployment is complete

Deployment name : CreateVm-canonical.ubuntu-24_04-lts-server-2...
Subscription : Azure for Students
Resource group : rg-vnet-vpn-lab
Start time : 11/04/2026, 11:34:42
Correlation ID : c38f9e6e-c747-400a-85fa-d67efb8c5c38

Deployment details

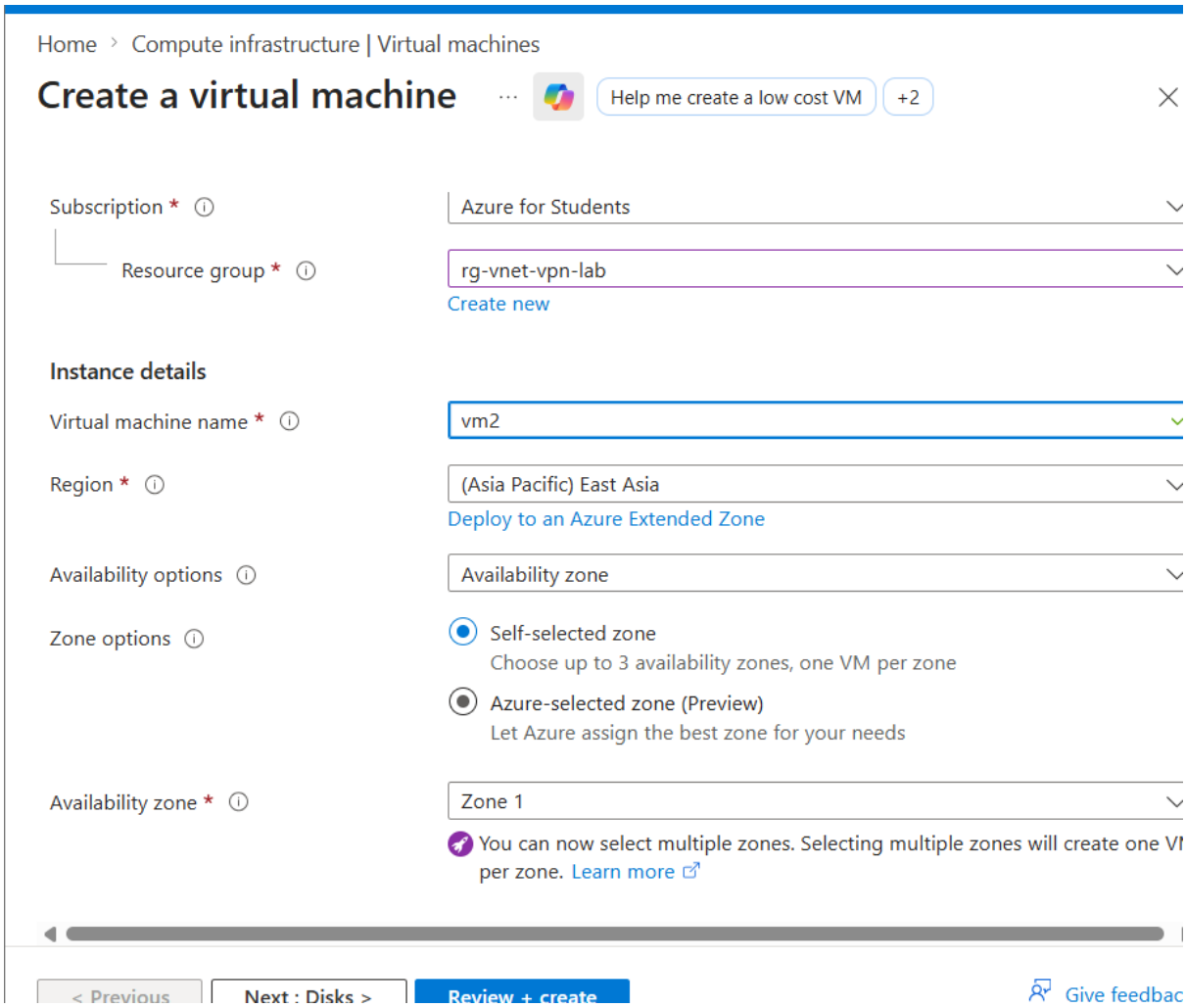
Next steps

Set up auto-shutdown Recommended
Monitor VM health, performance, and network dependencies Recommended
Run a script inside the virtual machine Recommended

Go to resource | Create another VM | Scale out your VM

*Add or remove favorites by pressing Ctrl+Shift+F*

**Step 6: Deploy Test VMs -VM1 AND VM2 Having VNET1 And VNET2 As virtual networks respectively**




**Step 7: Test Connectivity**

```
The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@vm2:~$ ping 10.0.1.4
PING 10.0.1.4 (10.0.1.4) 56(84) bytes of data.
64 bytes from 10.0.1.4: icmp_seq=1 ttl=64 time=1.63 ms
64 bytes from 10.0.1.4: icmp_seq=2 ttl=64 time=1.56 ms
64 bytes from 10.0.1.4: icmp_seq=3 ttl=64 time=1.66 ms
64 bytes from 10.0.1.4: icmp_seq=4 ttl=64 time=1.58 ms
64 bytes from 10.0.1.4: icmp_seq=5 ttl=64 time=1.84 ms
64 bytes from 10.0.1.4: icmp_seq=6 ttl=64 time=1.61 ms
64 bytes from 10.0.1.4: icmp_seq=7 ttl=64 time=3.61 ms
64 bytes from 10.0.1.4: icmp_seq=8 ttl=64 time=1.58 ms
^C
--- 10.0.1.4 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7012ms
rtt min/avg/max/mdev = 1.557/1.882/3.605/0.656 ms
azureuser@vm2:~$
```

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@vm1:~$ ping 10.1.1.4
PING 10.1.1.4 (10.1.1.4) 56(84) bytes of data.
64 bytes from 10.1.1.4: icmp_seq=1 ttl=64 time=2.37 ms
64 bytes from 10.1.1.4: icmp_seq=2 ttl=64 time=3.04 ms
64 bytes from 10.1.1.4: icmp_seq=3 ttl=64 time=1.72 ms
64 bytes from 10.1.1.4: icmp_seq=4 ttl=64 time=1.68 ms
64 bytes from 10.1.1.4: icmp_seq=5 ttl=64 time=1.69 ms
64 bytes from 10.1.1.4: icmp_seq=6 ttl=64 time=1.71 ms
64 bytes from 10.1.1.4: icmp_seq=7 ttl=64 time=2.45 ms
64 bytes from 10.1.1.4: icmp_seq=8 ttl=64 time=3.00 ms
64 bytes from 10.1.1.4: icmp_seq=9 ttl=64 time=1.68 ms
64 bytes from 10.1.1.4: icmp_seq=10 ttl=64 time=1.66 ms
64 bytes from 10.1.1.4: icmp_seq=11 ttl=64 time=1.69 ms
64 bytes from 10.1.1.4: icmp_seq=12 ttl=64 time=1.59 ms
^C
--- 10.1.1.4 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11018ms
rtt min/avg/max/mdev = 1.590/2.024/3.042/0.520 ms
azureuser@vm1:~$ client_loop: send disconnect: Connection reset
```

```
 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat Apr 11 06:47:52 UTC 2026

  System load:  0.0               Processes:             130
  Usage of /:   5.8% of 28.02GB   Users logged in:       0
  Memory usage: 32%               IPv4 address for eth0: 10.0.1.4
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Sat Apr 11 06:26:09 2026 from 157.39.210.52
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@vm1:~$
azureuser@vm1:~$ nano vm2_key.pem
azureuser@vm1:~$
```

```
azureuser@vm1: ~

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@vm1:~$ ping 10.1.1.4
PING 10.1.1.4 (10.1.1.4) 56(84) bytes of data.
64 bytes from 10.1.1.4: icmp_seq=1 ttl=64 time=2.37 ms
64 bytes from 10.1.1.4: icmp_seq=2 ttl=64 time=3.04 ms
64 bytes from 10.1.1.4: icmp_seq=3 ttl=64 time=1.72 ms
64 bytes from 10.1.1.4: icmp_seq=4 ttl=64 time=1.68 ms
64 bytes from 10.1.1.4: icmp_seq=5 ttl=64 time=1.69 ms
64 bytes from 10.1.1.4: icmp_seq=6 ttl=64 time=1.71 ms
64 bytes from 10.1.1.4: icmp_seq=7 ttl=64 time=2.45 ms
64 bytes from 10.1.1.4: icmp_seq=8 ttl=64 time=3.00 ms
64 bytes from 10.1.1.4: icmp_seq=9 ttl=64 time=1.68 ms
64 bytes from 10.1.1.4: icmp_seq=10 ttl=64 time=1.66 ms
64 bytes from 10.1.1.4: icmp_seq=11 ttl=64 time=1.69 ms
64 bytes from 10.1.1.4: icmp_seq=12 ttl=64 time=1.59 ms
^C
--- 10.1.1.4 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11018ms
rtt min/avg/max/mdev = 1.590/2.024/3.042/0.520 ms
azureuser@vm1:~$
```

```
azureuser@vm1:~$ whoami
azureuser
azureuser@vm1:~$ ssh -i vm2_key.pem azureuser@10.1.1.4
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1010-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat Apr 11 07:01:30 UTC 2026

  System load:  0.0               Processes:             126
  Usage of /:   5.7% of 28.02GB   Users logged in:       0
  Memory usage: 4%                IPv4 address for eth0: 10.1.1.4
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
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the exact distribution terms for each program are described in the
```